

Papers and Articles

Causes of death of wild birds of the family Fringillidae in Britain

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The provision of supplementary food for wild birds in gardens during the winter months is common in the UK, but it is possible that it may precipitate infectious diseases in the birds. This paper describes the results of postmortem examinations of 116 wild finches carried out over a period of four years. The two commonest causes of death in areas where high mortality had been reported were infections with the bacteria *Salmonella typhimurium* DT40 and *Escherichia coli* O86. Coccidia of the genera *Atoxoplasma* or *Isospora* were found in several of the birds but were considered to be incidental. Megabacteria were also identified in some of the birds, for the first time in flocks of wild birds in the UK, but they were not considered to be significant.

BIRDS of the order Passeriformes, family Fringillidae, usually referred to as finches, are small seed-eating birds with strong bills; the males have a well developed song and colourful plumage (Flegg 1990). Their natural habitats include hedgerows, woods, farmlands, orchards, commons and waste-lands, and seven species are included in the list of the 50 bird species most commonly recorded feeding in British gardens (Glue 1982). In order of frequency these are the chaffinch (*Fringilla coelebs*), greenfinch (*Carduelis chloris*), bullfinch (*Pyrrhula pyrrhula*), brambling (*F. montifringilla*), siskin (*C. spinus*), goldfinch (*C. carduelis*), and lesser redpoll (*Acanthis flammea*). Since the early 1960s, large numbers of greenfinches have been attracted into gardens by the provision of peanuts which can constitute a high proportion of the diet of the birds (Newton 1967). Siskins have also acquired the habit of feeding on peanuts in gardens (Spencer and Gush 1973) resulting in large flocks of siskins at some feeding stations. However the congregation of large numbers of birds in relatively small areas may increase their vulnerability to infectious diseases. This paper describes the causes of death of 116 finches from various parts of Scotland between 1994 and 1997.

Materials and methods

Birds examined

Highland Region 1994 and 1995. – In April and May 1994 the carcasses of 31 wild birds of the family Fringillidae were received from 10 sites around Inverness. Most of the sites were household

gardens in which supplementary food was provided for wild birds, especially in the form of peanuts, and to which large numbers of finches were attracted. The carcasses were often representative of a larger number of carcasses, and there were usually reports of other birds being seen alive but unwell, whose bodies were never found. A further 15 dead finches were received from six similar sites in April and May 1995, one of which had also been affected in 1994.

Strathclyde Region 1995-1996. – As part of a survey into the causes of death of wild birds undertaken between January 1995 and December 1996, the carcasses of 14 finches were examined. The majority were isolated incidents, but four were representative of larger numbers of birds found dead in May 1996 on two sites where wild birds were given supplementary food.

Grampian Region 1997. – As a result of reports of many ill and dead finches at bird feeding stations 56 finch carcasses were received from nine sites; 49 of them came from three sites. The great majority of the birds were greenfinches. Fifty-two of the 56 birds were received between January and April, and the remainder in May, June and July.

Examination of birds

A gross postmortem examination was carried out on all the birds, including an assessment of body condition based on the prominence of the sternum and the convexity or concavity of the breast muscles. Bacterial cultures were made from all the birds from the Highland and Strathclyde Regions, and from some of those from Grampian. Usually the liver and intestine, and sometimes other tissues, were inoculated on to Columbia agar (Difco) supplemented with 5 per cent citrated sheep blood, MacConkey agar without salt (Difco), Selenite F broth (Mast Laboratories) and deoxycholate citrate lactose sucrose agar (Difco), which were incubated aerobically at 37°C for 18 to 24 hours. Wet smears were prepared from the proventriculus and small intestine of all the birds from the Strathclyde and Grampian Regions, and examined microscopically for evidence of megabacteria, helminth eggs, and coccidial oocysts. Early in the investigation into the causes of mortality in the Grampian Region, impression smears were made from the liver and spleen, stained with Giemsa and with a modified Ziehl-Neelsen stain, and examined for coccidial merozoites and chlamydial elementary bodies, and wet preparations from the oesophagus were examined for the presence of motile protozoa. When appropriate, material was fixed in 10 per cent formol saline for histopathological examination.

Results

Highland Region 1994 and 1995

O86 colibacillosis. – A diagnosis of O86 colibacillosis was made in 27 birds in 1994 and 14 birds in 1995, mostly male and female

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TABLE 1: Numbers of carcasses of birds of different species received from three areas of Scotland between 1994 and 1997 and their diagnoses

Location and year	Diagnosis	Number of birds	Description
Highland 1994	O86 colibacillosis	27	14 male siskins 8 male greenfinches 3 female siskins 1 female greenfinch 1 male chaffinch
	Other or no diagnosis	4	2 female chaffinches 1 female greenfinch 1 male greenfinch
	Concurrent coccidia	2	1 female chaffinch 1 male greenfinch
	Concurrent megabacteria	1	1 male siskin
Highland 1995	O86 colibacillosis	14	7 male siskins 4 female siskins 1 male greenfinch 1 female greenfinch 1 male chaffinch
	Other or no diagnosis	1	1 female siskin
Strathclyde 1995-1996	O86 colibacillosis	4	2 male greenfinches 1 male chaffinch 1 female chaffinch
	Trauma	5	1 male bullfinch 1 male goldfinch 1 female goldfinch 1 male chaffinch 1 male greenfinch
	Other or no diagnosis	5	3 male greenfinches 2 female chaffinches
	Concurrent coccidia	5	1 male greenfinch 1 male goldfinch 1 female goldfinch 1 male bullfinch 1 female chaffinch
Grampian 1997	Salmonellosis	50	36 male greenfinches 13 female greenfinches 1 male chaffinch
	Trauma	2	1 male greenfinch 1 female chaffinch
	Other or no diagnosis	4	1 male bullfinch 2 female greenfinches 1 male greenfinch
	Concurrent megabacteria	15	9 male greenfinches 6 female greenfinches
	Concurrent coccidia	5	2 male greenfinches 2 female greenfinches 1 male bullfinch

siskins and male greenfinches which were in good to moderate body condition (Table 1). Many had accumulations of fragmented peanuts in the oesophagus, proventriculus or gizzard, sometimes in large quantities. In most of the remaining birds the gizzard contained only small particles of grit and either no food or only small quantities of seeds. The intestines either contained fluid which was discolored red, green, brown or black, or they were empty. No other significant gross findings were made. None of the birds showed evidence of trauma. Moderate or profuse growths of a non-lactose fermenting or late-lactose fermenting *Escherichia coli*, later identified as *E coli* O86:K61 (Foster and others 1998) were recovered from the intestines of all these birds, and from the livers of most of the birds. Histopathological examination did not reveal any consistent abnormalities, but postmortem autolysis precluded a detailed examination of the intestines. In one siskin, histopathology detected large numbers of organisms with the morphology of megabacteria on the epithelium of the proventriculus, and moderate numbers of coccidia were present in the superficial epithelium of the villi of the small intestine of a greenfinch. No reaction to the megabacteria or coccidia was visible, and there was no evidence of a systemic infection.

Other or no diagnosis. – Five birds were not considered to be typical of O86 colibacillosis because *E coli* O86 was either not isolated or was recovered only as a sparse growth. A female chaffinch, from which *E coli* O86 was isolated only in very small numbers,

had a very fatty liver and histopathological examination revealed small numbers of coccidial gamonts in the epithelium of the villi of the upper intestine. No diagnosis was made in a female chaffinch, the contents of whose digestive tract were normal, and which was *E coli* O86-negative. A male greenfinch with an empty digestive tract yielded only a sparse growth of *E coli* O86 from the intestine, and a female greenfinch which had fragments of peanut in the oesophagus and gizzard and abnormal intestinal contents was *E coli* O86-negative. *Listeria monocytogenes* was isolated in heavy growths from the intestine and liver of a female siskin; no fragments of peanut were found in this bird, but its terminal large intestine was distended by a mass of faeces and urates.

Strathclyde Region 1995-1996

O86 colibacillosis. – This condition was diagnosed in two greenfinches and two chaffinches received in May 1996 from two sites with a history similar to that of the Highland incidents. Many fragments of peanut were present in the pharynx, oesophagus and gizzard of the greenfinches, but only small seeds in the gizzard of the chaffinches. A profuse growth of *E coli* O86 was isolated from the liver and intestine of all four birds. No megabacteria were identified. A few unsporulated coccidial oocysts were found in smears from the upper intestine of one of the male greenfinches.

Trauma. – Traumatic injuries were found in five birds in good body condition. Cultures from the small intestine and liver of these birds did not yield *E coli* O86, but did yield sparse growths of a variety of bacteria such as *Enterobacter agglomerans*, *Serratia marcescens*, lactose-fermenting *E coli*, and *Enterococcus faecalis*. No megabacteria were found in any of these birds, but large numbers of coccidia were present in intestinal smears from the two goldfinches and the bullfinch; they were identified by in vitro sporulation as species of *Isospora* or *Atoxoplasma*.

Other or no diagnosis. – Starvation was considered to be the cause of death of two male greenfinches and a female chaffinch, which were thin, had no food in the digestive tract, and had discolored reddish-brown intestinal contents. Cultures for *E coli* O86 and salmonellae were negative. Profuse growths of *E faecalis* were isolated from the heart, liver and intestine of a female chaffinch with egg yolk peritonitis, in which a few unsporulated coccidial oocysts were also observed in intestinal smears. No diagnosis was made in an adult male greenfinch in which autolysis was advanced, although a profuse growth of *Yersinia enterocolitica* was recovered from a pharyngeal swab. No megabacteria were identified in any of the birds.

Grampian Region 1997

Salmonellosis. – Gross postmortem lesions consistent with salmonellosis were found in 48 greenfinches, and *Salmonella typhimurium* DT40 was isolated from the heart, liver and intestine of all the 19 birds from which cultures were made. *S typhimurium* DT40 was also isolated from the viscera of a greenfinch and a chaffinch in which no gross lesions were detected. Birds from seven of the nine sites were identified as salmonella-positive.

The body condition of the 49 greenfinches varied from good to poor, and many of the birds had food, predominantly fragments of peanuts, in the upper digestive tract. The commonest post-mortem lesion, found in 41 of them, was the presence of multiple, often coalescing, yellow nodules on the mucosa of the oesophagus, visible from the serosal surface and protruding into the lumen. These nodules measured 1 mm to 4 mm in diameter, and sometimes partially obstructed the oesophagus. Fragments of peanuts were frequently found adhering to the nodules. Hepatic lesions were present in 32 of the greenfinches, most often in the form of a swollen liver with multiple areas of fine focal necrosis 1 to 2 mm in diameter, but in some birds there was a perihepatitis or larger granulomata. In 28 of the greenfinches, the spleens were



pale and enlarged, measuring approximately 15 to 20 mm in length and 5 mm in diameter, with multiple yellow nodules 1 to 3 mm in diameter. A normal spleen would measure approximately 5 to 10 mm in length and 2 mm in diameter. Less frequently, a few pale nodules 2 to 3 mm in diameter were present in the oropharynx, choana, larynx and on the serosal surface of the gizzard. A purulent pericarditis was found in two greenfinches, and in one greenfinch the only postmortem abnormality was a purulent peritonitis.

Histopathology revealed a granulomatous hepatitis and splenitis, and the oesophageal lesions consisted of large necrotic masses surrounded by mixed inflammatory cells and superficial ulceration. No evidence of protozoa, fungi or viral inclusion bodies was found by the histopathological examination of liver, spleen or oesophagus.

No chlamydial elementary bodies or protozoal merozoites were identified in impression smears from livers or spleens, and no motile protozoa were identified in wet preparations from the oesophagus. Moderate to large numbers of megabacteria were detected in wet smears from the proventriculi of 12 of the greenfinches, and a few to many coccidial oocysts in four of them. On one occasion the oocysts were identified by *in vitro* sporulation as species of *Isospora* or *Atoxoplasma*.

Trauma. – Trauma was the cause of death of two birds, a male greenfinch which was the only bird received from the site, and a female chaffinch from a site on which multiple deaths from salmonellosis had been diagnosed. No coccidia were detected, but moderate numbers of megabacteria were found in the greenfinch. Neither *S typhimurium* DT40 nor *E coli* O86 was recovered.

Other or no diagnosis. – Lung congestion was the only abnormality found in two female greenfinches from a site on which salmonellosis had been diagnosed. No coccidial oocysts were found in either bird, but one had a few megabacteria and the other had many megabacteria. A sparse growth of a lactose-fermenting *E coli* was recovered from the liver and small intestine of one bird, and cultures from the second bird were sterile. Starvation was diagnosed in a thin male greenfinch (the only bird received from the site) from which moderate growths of *Enterobacter agglomerans* were recovered. No coccidia or megabacteria were detected. Starvation and kidney failure was diagnosed in a male bullfinch submitted from a site on which there had been many deaths from salmonellosis in greenfinches. No megabacteria were found, a few coccidial oocysts were detected in the upper intestine, and *Staphylococcus xylosum* was cultured from the small intestine and liver, and a lactose-fermenting *E coli* from the kidney.

Discussion

Mortality associated with *E coli* O86:K61 was the major problem diagnosed in finches in the Highland Region in 1994 and 1995, being confirmed in 87 per cent and 93 per cent of the carcasses received in the two years. The condition was also diagnosed in Strathclyde in 1996, at the only two sites on which multiple deaths were reported. The biochemical properties and toxigenicity of the *E coli* O86 isolated from the finches in 1994 and 1995 have been described by Foster and others (1998).

All of the strains of *E coli* O86 tested were positive for the *eae* gene and produced cytolethal distending toxin (CLDT) (Foster and others 1998), indicating the potential of the organism to cause disease in the finches. In human beings *E coli* O86:K61 is associated with outbreaks of diarrhoea in infants (Ewing 1986).

Although *E coli* O86 was isolated from a high proportion of the birds from the Highland Region, it was not found in most of the finches from Strathclyde, except in those from sites where multiple deaths had been reported. There appears to be an association between multiple deaths on a site or in an area and the recovery of *E coli* O86. The organism does not appear to be widespread in other species of wild birds, because it was not cultured from the carcasses of 101 non-finch wild birds which died from a variety of different causes between 1995 and 1996 (T. Pennycott, unpub-

lished observations). However, *E coli* O86 has been isolated from wild sparrows (*Passer* species) in Poland (Pawiak and others 1991), and that study concluded that *E coli* could contribute to mortality in nestlings.

Salmonellosis was the major diagnosis in the Grampian birds, and affected 89 per cent of the finches received. *S typhimurium* has been recovered from finches and other passerines such as the house sparrow (*P domesticus*) on many occasions (Wilson and McDonald 1967, McDonald and Brown 1974, Laing 1990). More recently, Kirkwood and others (1995) reported a number of deaths at bird tables of greenfinches and house sparrows, and other birds such as siskins, tree sparrows (*Passer montanus*) and blue tits (*Parus caeruleus*), in which *S typhimurium* was believed to be responsible. A similar incident involving greenfinches and a chaffinch was described by Routh and Sleeman (1995).

A common feature of such incidents is the repeated congregation of large numbers of susceptible birds at restricted sites, such as feeding stations set up to provide food for wild birds in winter, or industrial premises providing overnight roosts. It is clear that pathogenic bacteria such as *S typhimurium* can readily spread through flocks under these conditions. Although several different phage types of *S typhimurium* have been isolated from finches and sparrows, phage type 40 appears to be the most important. This phage type (then referred to as U165) was one of three reported to be common in incidents of wild bird mortality in the south of England every spring (McDonald and Brown 1974), and it was found in greenfinches in different areas of England in the spring of 1995 (J. P. Holmes, A. Routh, personal communications) and was responsible for the heavy mortality in finches in the Grampian Region of Scotland in 1997.

A notable feature of salmonellosis in finches in Grampian was the fact that oesophageal lesions were recorded in 84 per cent of the birds with lesions of salmonellosis. Oesophageal lesions in greenfinches were also described by Kirkwood and others (1995), and Wilson and McDonald (1967) reported oesophageal ulceration with pus formation in greenfinches.

Outbreaks of salmonellosis in greenfinches are therefore consistent in terms of the time of year at which they occur, the phage type of *S typhimurium* involved, and the gross postmortem findings. It seems unlikely that these birds are acting as indicators of environmental contamination from another source, but rather that the greenfinches themselves are the source of the infection, which then sometimes affects other garden birds.

Of the carcasses examined, 72 per cent were male birds and 28 per cent were female, largely because 76 per cent of the carcasses with salmonellosis or colibacillosis were male and only 24 per cent were female. However, among the birds in which another or no diagnosis was made, 52 per cent were male and 48 per cent female. It is unclear whether the high proportion of males with salmonellosis or colibacillosis was due to there being a high percentage of males in the winter population of greenfinches and siskins, as can occur in some flocks of immigrant chaffinches (Newton 1972), whether the more dominant males spend more time at the feeding stations, or whether there may be another explanation.

Megabacteria were present on the epithelium of the proventriculus of a male siskin, nine male greenfinches and six female greenfinches. This organism, of uncertain classification, has been associated with mortality in several species of psittacine and passerine birds (Filippich 1994), including wild goldfinches and siskins in the Netherlands (Cornelissen 1993) and wild goldfinches in Australia (Filippich 1994). This is the first report of megabacteria in flocks of wild birds in the UK. An examination for megabacteria did not form part of the standard procedure used for birds from the Highland Region, and the organism may therefore have been present in more birds than reported. However, no associated inflammatory or degenerative changes were apparent in the proventriculus either by gross or histopathological examination, and the presence of megabacteria was considered to be incidental. The mean dimensions of the megabacteria recovered from one of the greenfinches were 32 µm × 2.7 µm, similar to the dimensions of megabacteria recovered from other avian species (T. Pennycott, unpublished observations).



Coccidia with the morphology of *Isospora* or *Atoxoplasma* species were observed in wet preparations from the intestines of 10 of 70 birds from Strathclyde and Grampian, five greenfinches, two goldfinches, two bullfinches and one chaffinch. Coccidial stages presumed to be *Isospora* or *Atoxoplasma* species were also observed on histopathology of the upper intestine of a greenfinch and a chaffinch from the Highland Region. Coccidia of the genera *Isospora* and *Atoxoplasma* both produce oocysts which are passed in the faeces and look very similar, because after sporulation they both contain two sporocysts each with four sporozoites; as a result there has been confusion about the identity of different coccidia found in birds (Levine 1982a, b). Such coccidia are frequently found in finches; for example Anwar (1966) recovered *Isospora* or *Atoxoplasma* from the faeces of 28 greenfinches and four chaffinches which had been captured and held in captivity.

The presence of *Isospora* in the intestine does not appear to be associated with pathological changes (Arnall and Keymer 1975), but *Atoxoplasma*, which has developmental stages in the tissues or blood, has been associated with weight loss and mortality in captive-bred goldfinches (Sironi 1994), bullfinches (McNamee and others 1995) and greenfinches (Cooper and others 1989), and in recently imported black siskins (Rossi and other 1996) in which postmortem lesions such as hepatomegaly and splenomegaly may be observed.

In this study, it was not certain whether the coccidia in the 12 finches were species of *Isospora* or *Atoxoplasma*. However, there were no lesions in the liver or spleen of the birds with coccidia from the Highland and Strathclyde Regions, and none in one of the five birds with coccidia from Grampian. The other four birds from Grampian had enlarged livers and spleens but these changes were ascribed to salmonellosis. Whether the coccidia were *Isospora* or *Atoxoplasma* species, they were considered to be incidental findings.

It is beyond the scope of this paper to discuss the merits of providing additional food for wild birds in winter. However, some of the potential disadvantages have been illustrated by the incidents described. Encouraging the congregation of large numbers of birds in a small area, sometimes for prolonged periods, is likely to result in heavy microbiological contamination of the feeding stations. The small amount of feeding space available to each bird leads to aggressive behaviour and male dominance at the feeding stations (Newton 1972), stressing the birds and increasing their susceptibility to infectious diseases, including those described in this study and others such as chlamydiosis (Simpson and Bevan 1989). These infectious diseases may then infect other species of wild birds in the vicinity, and may also pose a threat to human health. The two major organisms identified, *S typhimurium* DT40 and *E coli* O86, are both potentially zoonotic, as is *Chlamydia psittaci* (Cooper 1990) and other bacteria isolated from the finches. Potentially zoonotic strains of *E coli* have also been isolated from feral pigeons (Dell'Omo and others 1998). Practices which encourage large numbers of wild birds to visit gardens may therefore increase the hazards to human health, and suitable hygienic precautions should be taken when cleaning wild bird feeding stations and utensils, and if sick or dead wild birds have to be handled.

In addition to posing a threat to other wild birds and people, infected populations of wild birds also form a reservoir of infection for domestic livestock and companion animals. For example, this study has highlighted the significance of *S typhimurium* DT40 in finches, but the organism has also been recovered from people, horses, gamebirds, and other farmed livestock. Of 65 incidents of *S typhimurium* DT40 reported to the Epidemiology Department of the Central Veterinary Laboratory between 1990 and 1996, 11 involved finches, 19 were in pheasants or partridges (including 15 in one year), 13 were in horses, seven were recorded in ducks or geese, four were in chickens, four in cattle, and the remainder in other miscellaneous or unidentified avian or mammalian species (C. Byrne, personal communication).

The consequences of feeding large numbers of wild birds for prolonged periods in the winter months need to be more fully determined. If the health of wild bird populations is being threatened by such practices, or the health of people, livestock or com-

panion animals is being jeopardised, then alternative ways of helping wild birds in winter may be required.

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Abstract

Postsurgical changes in the cruciate deficient canine stifle joint

FIFTY-EIGHT dogs with osteoarthritis of the stifle joint secondary to cranial cruciate ligament deficiency were examined before, and one-and-a-half, seven and 13 months after surgical treatment. Initially, periarticular thickening and the degree of crepitus were significantly correlated with the duration of the disease. During the period of the study the degree of quadriceps atrophy decreased significantly, although it remained abnormal, suggesting that a measure of quadriceps atrophy may be a useful indication of the long-term outcome of the disease.

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